Accurate Estimation of Small Effects: Illustration Through Air Pollution and Health

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Abstract

This paper identifies design parameters that can lead to inaccurate estimates of small effects. Low statistical power not only makes such effects difficult to detect but resulting significant estimates typically exaggerate effect sizes. Through the literature on short-term health effects of air pollution, we explore the prevalence and policy implications of this issue. While some designs seem robust, others likely suffer from substantial exaggeration. Real-data simulations highlight key drivers often hidden behind reassuring features such as large sample sizes: the number of exogenous shocks, instrument strength, and outcome distribution. We discuss approaches to evaluate and mitigate exaggeration in non-experimental studies.

Keywords: Causal Inference, Exaggeration, Statistical Power, Acute Health Effects, Air Pollution, Simulations

Project's website: https://vincentbagilet.github.io/inference_pollution

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Accurate estimates are central to credible economic analysis and public policy. Empirical findings routinely inform regulatory decisions, benefit-cost analyses, and resource allocation, so inaccurate estimates can lead to misguided policies and substantial welfare losses. Yet accurate estimation is challenging. In most applied studies, if not the main effect, at least some effects of interest are relatively small, for instance when implementing heterogeneity analyses or exploring impacts on secondary outcomes. They can thus be difficult to capture, even when the sample is large and identification is valid. When underlying effects are small, concrete design choices become decisive for the reliability of the resulting estimates. In particular, they can determine whether a study has enough statistical power not only to detect but also to accurately estimate effects of interest.

The present paper identifies tangible design parameters that can lead to low power and consequently to inaccurate estimates of relatively small effects. It points to concrete factors that we, as applied researchers, can directly act on to reduce this risk in non-experimental studies and explores the consequences of inaccurate estimates for public policy, through the lens of the literature on the short-term health effects of air pollution. This literature provides a particularly well-suited setting for this analysis. Its studies vary widely in identification strategies, effect sizes, population of interest, and sample sizes considered. This heterogeneity enables us to evaluate, within a single framework, the many parameters and settings regularly encountered in applied economics, yielding insights that generalize well beyond this specific literature.

To identify tangible drivers of inaccurate estimation, we proceed in four steps. First, we illustrate through an example how low statistical power causes statistically significant estimates to exaggerate true effect sizes. Low power does not prevent obtaining a significant estimate, but such estimates typically exaggerate the true effect size and, when combined with selection on significance, so do published estimates (Gelman and Tuerlinckx 2000, Ioannidis 2008, Gelman and Carlin 2014). Power is the study-level driver of estimation biases resulting from selective publication. Second, we document substantial heterogeneity in exaggeration across the literature on the short-term health effects of air pollution and assess implications for policy design. Third, we systematically identify drivers through Monte Carlo simulations using data from the US National Morbidity, Mortality, and Air Pollution Study (Samet et al. 2000), replicating common identification strategies: standard regression, reduced-form, instrumental variable, and regression discontinuity designs. Finally, we propose a workflow for evaluating exaggeration risks when conducting a non-experimental study.

In the meta-analysis, we analyze 2,692 estimates of the short-term health effects of air pollution from a unique corpus of 668 epidemiological studies and 36 articles relying on causal inference methods. This analysis reveals substantial heterogeneity in exaggeration risk. While some studies appear well-powered, many face severe limitations. Strikingly, we find that less precise studies systematically report larger standardized effect sizes. Only 11% of causal inference studies and 42% of epidemiology studies would be adequately powered to detect an effect half the size of their estimate, for the conventional 80% power threshold. Benchmarked against OLS estimates, half of the instrumental variable designs would lead to an exaggeration of the effects by a factor of at least 4.5, and a quarter by at least 8.2. Studies exploiting rare events like air quality alerts or transportation strikes can also face severe power constraints. These findings have policy implications as academic estimates directly inform the monetization of health benefits in US Environmental Protection Agency (EPA) regulatory impact analyses. For exaggeration ratios comparable to the ones we document, annual misperceptions of net benefits could reach $10 to $35 billion. While air pollution's large benefit-cost ratio preserves positive net benefits, exaggeration of this magnitude could misallocate resources across policies, and in regulatory contexts with tighter benefit-cost margins, could reverse policy decisions. The accuracy and precision of these estimates are under direct scrutiny: in January 2026, the EPA announced it would no longer monetize the health benefits of reducing PM2.5 and ozone in Clean Air Act rulemakings, citing excessive uncertainty in existing estimates (U.S. EPA 2026). This decision further underlines the importance of understanding what drives imprecision in these estimates and should invite to improve measurement rather than to abandon such critical quantification altogether.

Our simulations identify concrete design features behind these patterns that extend to most settings, beyond air pollution and health. Exaggeration often arises even for large sample sizes and for effect sizes typically observed in the literature. It also increases, as expected, when these quantities decrease. Beyond this familiar pattern, we show the importance of three less straightforward drivers. First, relying on rare exogenous shocks can produce greatly inflated estimates. Such shocks can represent less than 1% of the observations, for instance in studies leveraging public transportation strikes or thermal inversions, often leading to large exaggeration ratios. Second, substantial exaggeration can arise when the instrument only explains a limited portion of the variation in the explanatory variable of interest, even when F-statistics are large. Finally, exaggeration is affected by the count of cases of the outcome. When this count is low, as for instance for hospital admissions or deaths among the elderly or children, estimated effects can be substantially exaggerated. Many of these situations are common in non-experimental research, yet they can be hidden behind otherwise reassuring study features such as large sample sizes.

This paper makes three main contributions. First, it contributes to the literature on the bias of published estimates arising from selective publication. The estimation biases resulting from selective publication, whose detection and correction are formalized for instance in Andrews and Kasy (2019), are conditional on low power and, at the study level, driven by it. Adequately powered studies do not lead to substantial exaggeration, even in the presence of selection on significance. The literature on publication bias and associated estimation biases typically focuses on one of the two ingredients for exaggeration, selective publication, operating at the level of a body of literature to detect and correct for selection. What drives low statistical power in practice, the second ingredient, remains largely uncharacterized. The present paper operates at the level of individual studies and identifies the design features that determine whether a study is prone to exaggeration. Existing meta-analyses document serious power issues in economics but typically do not identify their determinants (Ioannidis, Stanley and Doucouliagos 2017, Ferraro and Shukla 2020, Young 2022, Arel-Bundock et al. 2024). Outside of meta-analyses, to our knowledge, only three papers systematically examine exaggeration drivers (Schell, Griffin and Morral 2018, Griffin et al. 2021, Black et al. 2022), all focusing on difference-in-differences event-study designs. We complement this work by studying the concrete drivers of exaggeration in a wide array of research designs: standard regression, reduced-form, instrumental variable, and regression discontinuity designs. We identify actionable design parameters that researchers can evaluate in their own work to assess exaggeration risk. In Bagilet (2026), a companion paper, we show that causal identification strategies can amplify exaggeration by limiting the amount of variation used for identification. The present paper complements this theoretical point with concrete, actionable guidance and by documenting the policy consequences of exaggeration.

Second, we contribute to the literature on the replicability and credibility of empirical findings (Button et al. 2013, Open Science Collaboration 2015, Christensen and Miguel 2018, Camerer et al. 2016, Brodeur, Cook and Heyes 2020, Brodeur, Mikola and Cook 2024). We strive to put statistical power at the center of non-experimental analyses, as it is critical even after a statistically significant estimate has been obtained. We propose reporting power calculations alongside published results. To this end, the present paper is built so that it can be used as a template that can be adapted to a wide range of settings to assess exaggeration risk in individual studies. The section analyzing the literature provides an example of how one can conduct a retrospective power analysis, while the simulations illustrate how real or simulated data can be used to identify potential sources of exaggeration before undertaking a study. [1]

[1] We used literate programming to make all replication and supplementary materials accessible and

Finally, we contribute to the literature on the acute health effects of air pollution by systematically evaluating the reliability of published estimates across research designs and analyzing implications for regulatory policy. This literature is remarkably vast, with hundreds of studies published on this topic. It also directly informs environmental policy addressing a major public health concern. However, typically small underlying effect sizes and coarse data, with observations generally at the city-day level, make it particularly prone to exaggeration and power issues. We document publication bias, low power, and exaggeration in this literature, show that certain designs can be more subject to these issues, and describe how exaggerated health estimates affect the air pollution regulatory process.

1 Background on Statistical Power and Exaggeration

Gelman and Tuerlinckx (2000) and Gelman and Carlin (2014) point out that statistically significant estimates suffer from a winner's curse in under-powered studies. These estimates can largely exaggerate true effect sizes or can even be of the opposite sign. In this section, we illustrate these two points through a simple simulation exercise. For clarity of exposition, this exercise builds on a short-term health effect of air pollution example but can easily be translated to more general settings.

1.1 Illustrative Setting

Let's simulate an "ideal" experiment in which a mad scientist is able to randomly increase the concentration of fine particulate matter ($PM_{2.5}$) to estimate the short-term effects of air pollution on daily non-accidental mortality. The experiment takes place in a major city over the 366 days of a leap year. The scientist increases the concentration of particulate matter by 10 μg/m^3–a large shock equivalent to a one standard deviation increase in the concentration of $PM_{2.5}$. Concretely, the scientist implements a complete experiment where they randomly allocate half of the days to the treatment group and the other half to the control group. They then measure the treatment effect of the intervention by computing the average difference in means between treated and control outcomes. They find a treatment effect of 4 additional deaths that is statistically significant at the 5% level. The statistical significance of the estimate fulfills the scientist's expectations.

Contrary to the scientist, we know the true effect of the experiment since we created the data. Table 1 displays the pair of potential outcomes of each day, $Y_i(T_i = 0)$ and

easily adaptable *via* the project's website.

$Y_i(T_i = 1)$. $Y_i(T_i)$ represents the daily count of non-accidental deaths and T_i the treatment assignment, equal to 1 when unit i is treated and 0 otherwise. We first simulate the daily non-accidental mortality counts in the absence of treatment (i.e., the $Y(0)$ column of Table 1), by drawing 366 observations from a negative binomial distribution with a mean of 106 and a variance of 402. We choose these parameters to approximate the distribution of non-accidental mortality counts in a large European city. We then define the counterfactual distribution of mortality by adding the treatment effect, drawn from a Poisson distribution (i.e., the $Y(1)$ column of Table 1). We choose its parameter to increase the number of deaths by 1 on average[2].

Table 1: Science Table of the Experiment.

Day Index	$Y_i(0)$	$Y_i(1)$	τ_i	T_i	Y_i^{obs}
1	122	124	+2	1	124
2	94	96	+2	1	96
3	96	98	+2	0	96
⋮	⋮	⋮	⋮	⋮	
364	96	97	+1	0	96
365	98	98	+0	0	98
366	143	144	+1	1	144

Notes: This table displays the potential outcomes, the unit-level treatment effect, the treatment status and the observed daily number of non-accidental deaths for 6 of the 366 daily units in the scientist's experiment.

Following the fundamental problem of causal inference, the daily count of deaths the scientist observes is given by the equation: $Y_i^{obs} = T_i \times Y_i(1) + (1 - T_i) \times Y_i(0)$. Considering that the assignment of the treatment was random, how can the statistically significant estimate found by the scientist be 4 times larger than the true treatment effect size? Replicating the experiment a large number of times explains this apparently puzzling result.

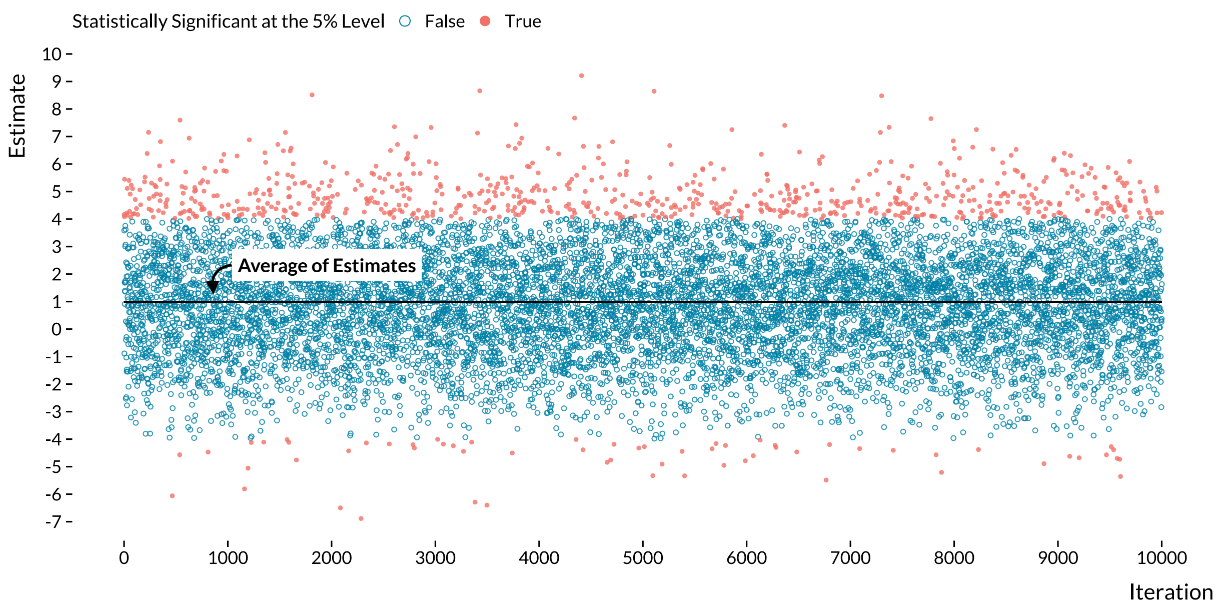
1.2 Defining Statistical Power and Exaggeration

Figure 1 plots the estimates of 10,000 iterations of the experiment. Even if there is a large variation in the effect size of estimates, their average is reassuringly equal to the true treatment effect of 1 additional death. We can however note that estimates close to

[2]In relative terms, the treatment effect size we set represents a 1% increase in the health outcome. The magnitude of this hypothetical effect is larger than the one found in a recent and large-scale study based on 625 cities. Liu et al. (2019) estimated that a 10 $\mu g/m^3$ increase in $PM_{2.5}$ concentration was associated with a 0.68% (95% CI, 0.59 to 0.77) relative increase in daily all-causes mortality.

the true effect would not be statistically significant at the 5% level. In a world without publication bias, several replications of this experiment would recover the true treatment effect, on average. Yet, despite recent changes in scientific practices and editorial policies, non-statistically significant estimates are less valued and replication exercises remain rare (Brodeur, Cook and Heyes 2020). In a world with publication bias, statistically significant estimates are more likely to be made public. Out of the 10,000 simulation estimates, about 800 are statistically significant at the 5% level. The *statistical power* of the experiment, which is the probability to reject the null hypothesis when there is actually an effect, is estimated to be 8%. The scientist was therefore lucky to get a statistically significant estimate.

Figure 1: Replicating 10,000 Times the Experiment.



Notes: Each dot represents a point estimate of one of the 10,000 iterations of the randomized experiment run by the mad scientist. Red dots are statistically significant at the 5% level while blue dots are not. The black solid line represents the average of estimates, equal to the true average effect of 1 additional death.

With such a low statistical power, statistically significant estimates are however not informative of the treatment of interest. Two metrics, the average type M (magnitude) error and the probability to make a type S (sign) error help assess the negative consequences of a lack of statistical power. The exaggeration ratio, or expected type M error, is defined as the ratio of the absolute values of the statistically significant estimates over the true effect size (Gelman and Carlin 2014). In the present case, with an estimated statistical power of 8%, the scientist could expect their statistically significant estimates to be inflated on average by a factor of 5. We also notice in Figure 1 that a non-negligible fraction of statistically significant estimates are of the wrong sign: this proportion is the probability of making a type S error (Gelman and Tuerlinckx 2000). In this experiment, a statistically significant estimate has an 8% probability of being of the wrong sign.

Formally, the statistical power of a test is the probability of rejecting the null hypothesis $H_0 : \beta = 0$, where β is the true effect of the estimand of interest. For $\hat{\beta}$, a normally distributed unbiased estimator of β with standard error σ, the power of the null hypothesis test at the 5% level is equal to $\Phi\left(-1.96 - \frac{\beta}{\sigma}\right) + 1 - \Phi\left(1.96 - \frac{\beta}{\sigma}\right)$, where Φ is the cumulative distribution function of the standard normal distribution. It increases with β, the true value of the effect and with the precision of the estimator, *i.e.*, when σ decreases. The exaggeration ratio is $\mathbb{E}\left(\frac{|\hat{\beta}|}{|\beta|} \,\middle|\, \beta, \sigma, |\hat{\beta}|/\sigma > 1.96\right)$ and the probability to make a type S error is given by $\Pr\left(\frac{\hat{\beta}}{\beta} < 0 \,\middle|\, \beta, \sigma, |\hat{\beta}|/\sigma > 1.96\right)$. Zwet and Cator (2021) and Lu, Qiu and Deng (2019) derive closed-form expressions for these quantities. They show that both the exaggeration ratio and the probability of type S error decrease with β, with the precision of the estimator and thus with statistical power.

1.3 Exaggeration of Published Estimates in Economics

Exaggeration of published estimates arises from the combination of publication bias and low statistical power. Without publication bias, all results would be published and estimates would be noisy but unbiased on average. Selection on significance is well documented in economics (Rosenthal 1979, Brodeur et al. 2016, Andrews and Kasy 2019, Vivalt 2019, Abadie 2020, Brodeur, Cook and Heyes 2020). Importantly, publication bias extends beyond significance: large or surprising results may also be favored, further amplifying exaggeration when power is limited (Andrews, Kitagawa and McCloskey 2024). Conversely, when statistical power is high, significant estimates closely approximate true effects and publication bias has little scope to distort them. Ioannidis, Stanley and Doucouliagos (2017) estimates that nearly 80% of results across a wide array of economics literatures are exaggerated by a factor of two or more. The two ingredients leading to exaggerated published estimates are thus present in the economics literature. In the following sections, we explore exaggeration and power issues in the literature on the short-term health effects of air pollution, complementing prior assessments of exaggeration in economics through a focus on causal identification approaches. This quantification enables us to document the potential impacts of exaggeration on the design of public policies.

2 Observed Exaggeration and Implications for Policy

From extreme events such as the London Fog of 1952 to the development of sophisticated time-series analyses, a vast epidemiology literature of more than 600 studies has established that air pollution induces adverse health effects over the short term. Increases in the concentration of several ambient air pollutants have been found to be associated with increases in daily mortality and emergency admissions for respiratory and cardiovascular causes (Schwartz 1994, Samet et al. 2000, Le Tertre et al. 2002, Bell, Samet and Dominici 2004, Liu et al. 2019). With this objective in mind, researchers in economics and epidemiology have recently used causal inference methods to improve on the standard epidemiology literature that relied on associations (Dominici and Zigler 2017, Bind 2019). Newly obtained results confirm the short-term health effects of air pollution (Schwartz et al. 2015, Schwartz, Fong and Zanobetti 2018, Deryugina et al. 2019). Based on these results, environmental protection and public health agencies design policies such as air quality alerts to mitigate the burden of air pollution. Accurate estimates of these effects are therefore crucial as they are directly used to implement and update policies to address this major public health issue.

This section first analyses this literature through the prism of statistical power and exaggeration. It describes how we ran retrospective analyses of the causal inference and standard epidemiology literatures as well as the necessary steps to make this analysis easily reproducible in other literatures or for individual studies. Then, we explore the impacts of exaggeration on the design of concrete public policies through the case of air pollution mitigation policies in the United States. While the analysis focuses on a specific topic, the intuitions extend to other settings where academic estimates inform policy.

2.1 Approach to the Literature Analysis

The formulas for power, exaggeration ratio, and type S error described in the previous section all depend on the true magnitude of the estimand, which is never observed. We address this with retrospective power analyses. The core question they address is: would the design of a study be reliable enough to retrieve the true effect if it were in fact smaller than the obtained estimate? A retrospective power analysis can be considered as a thought-experiment in which we would exactly replicate the study many times under the assumption that the true effect is different from the observed estimate. The reasoning is analogous to the analysis implemented in the previous section. Concretely, one can draw many estimates from the hypothetical asymptotic distribution of the estimator, a normal distribution centered on a hypothesized true effect with standard deviation equal to the standard error obtained in the study (Gelman and Carlin 2014). Statistical power is the proportion of sampled estimates that are statistically significant. The exaggeration ratio is the average ratio of the absolute values of significant estimates over the assumed true effect. The type S error rate is the proportion of significant estimates with the opposite sign of the true value. In our analysis, we use the `R` package `retrodesign` described in Timm, Gelman and Carlin (2019), which implements closed-form analogues of these simulations (Lu, Qiu and Deng 2019).

To get a general overview of power issues in the causal inference and standard epidemiology literatures, we first carry out a simple retrospective analysis for each study. These computations rely on hypothesized true effect sizes. Yet, since treatments and outcomes vary between studies in this literature, it is not possible to make general aggregated assumptions on true effect sizes. We need to consider specific hypothesized true effect sizes for each study. Since Ioannidis, Stanley and Doucouliagos (2017) and Ferraro and Shukla (2020) find a typical exaggeration of two in the economics and environmental economics literatures, we first report the proportion of studies that would have a design reliable enough to retrieve an effect size equal to half of the obtained estimate. We do not claim that the hypothesized effect size is the true effect size but rather wonder whether the design of the study would allow capturing an effect of this magnitude. A well-designed study should be able to detect a range of plausible effect sizes that are smaller than the observed estimate. We highlight results for such a hypothetical true effect but display power and exaggeration curves for a broad range of plausible effect sizes in Figures 2 and 5. Since these hypothetical values are constructed from potentially exaggerated published estimates, our results constitute a lower bound on actual exaggeration. To complement this approach, for a subset of studies, we use meta-analysis estimates from Shah et al. (2015) as alternative true effect sizes.

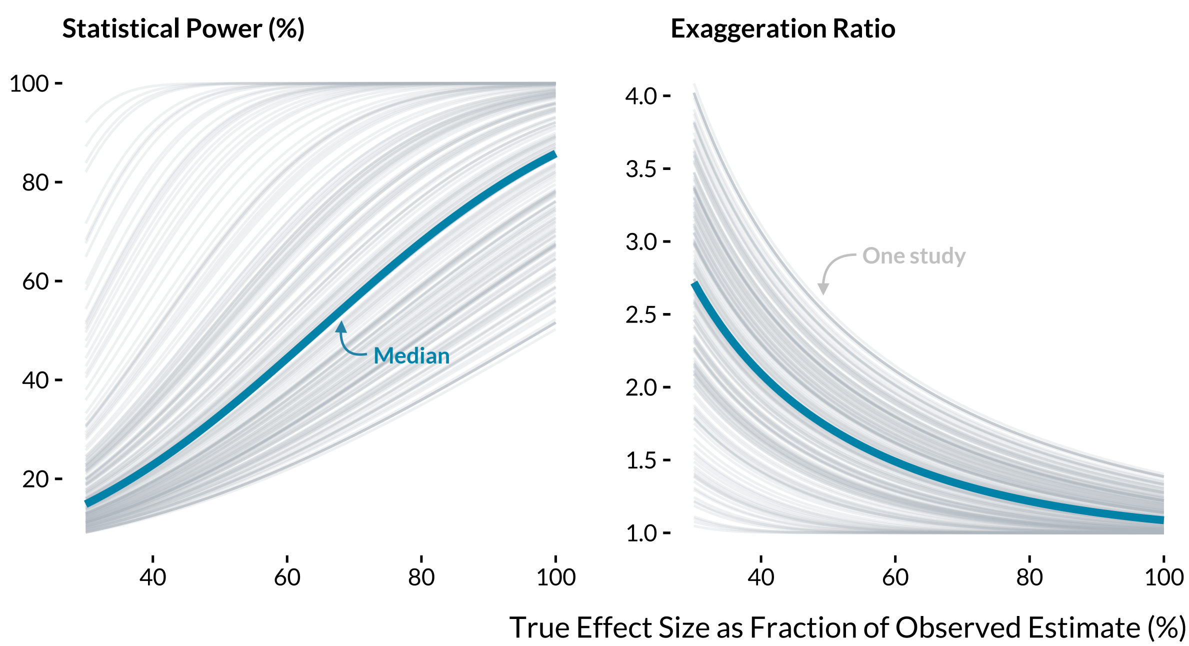
2.2 Exaggeration in the Causal Inference Literature

Through a thorough search strategy on Google Scholar, IDEAS, and PubMed, we retrieved all studies that (i) focus on the short-term health effects of air pollution on mortality and morbidity outcomes, and (ii) rely on a causal inference method.[3] Appendix A displays the list of the 36 articles that match the search criteria. For each study, we manually retrieve the method used by the authors, the health outcome and air pollutant they consider, the point estimate and the standard error of the main specifications.

[3]The very recent literature on the effects of air pollution on COVID-19 health outcomes is excluded to gather a relatively homogeneous corpus of studies.

To evaluate potential statistical power issues in this literature, we implement retrospective power calculations, considering a variety of hypothetical true effect sizes. Figure 2 plots the power and exaggeration curves for 186 specifications whose results are statistically significant at the 5% level. For instance, if the true effect size of each study were equal to half of the obtained estimate, the median power would be 33% and the median exaggeration ratio would be 1.7. Only 11% of studies would have a power greater than 80%. Figure 2 also shows that there is a wide heterogeneity in statistical power issues among studies. Some are estimated to be relatively well powered while others exhibit large exaggeration ratios. For instance, one quarter of studies would, on average, exaggerate the true effect size by a factor greater than 2. This pattern may help explain the very large effect sizes sometimes observed in the causal inference literature.

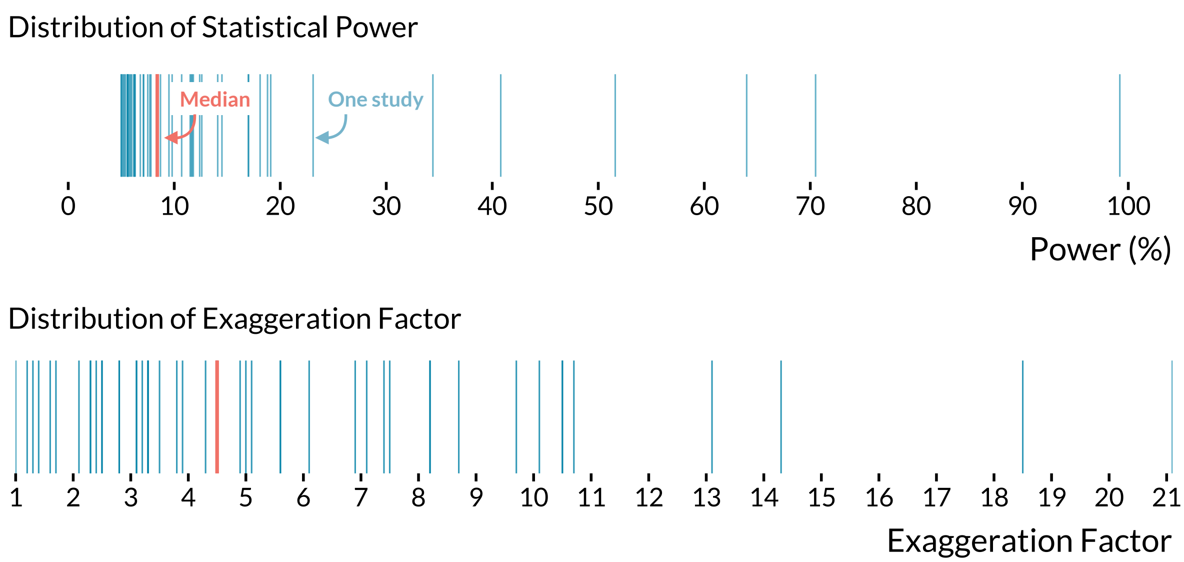
Figure 2: Statistical Power and Exaggeration Curves of Causal Inference Studies.



Notes: Each gray line is a power curve or an exaggeration curve of a statistically significant result published in the causal inference literature. The blue lines are the median values. For visual clarity, we drop results for which exaggeration ratios were too large.

Then, for the 49 instrumental variable results that are statistically significant and that report the corresponding naive regression results, we evaluate whether the 2SLS designs would recover an effect close to that of the naive estimate. We do not claim that this naive estimate would be the true effect but would expect a 2SLS design to be able to recover an effect of this magnitude. Figure 3 displays the distribution of the resulting statistical power and the average exaggeration ratio. The median estimated power is 8.4%, which translates into very large exaggeration ratios: half of the studies would exaggerate a true effect of the size of the OLS estimate by a factor of at least 4.5. Some designs are associated with extremely large exaggeration ratios, exceeding 8.2 for a quarter of studies. This heterogeneity is both reassuring and concerning: while some designs appear adequately powered, a substantial share of studies might suffer from severe exaggeration. Such inflation of statistically significant estimates could explain part of the gap between the standard and causal literatures. This discrepancy could also reflect a combination of omitted variable bias and attenuation bias from classical measurement error in air pollution exposure, or the fact that the causal estimands targeted by both strategies differ when treatment effects are heterogeneous. These explanations are not mutually exclusive, and the lack of power and inability of instrumental variables to recover smaller effect sizes remain concerning.

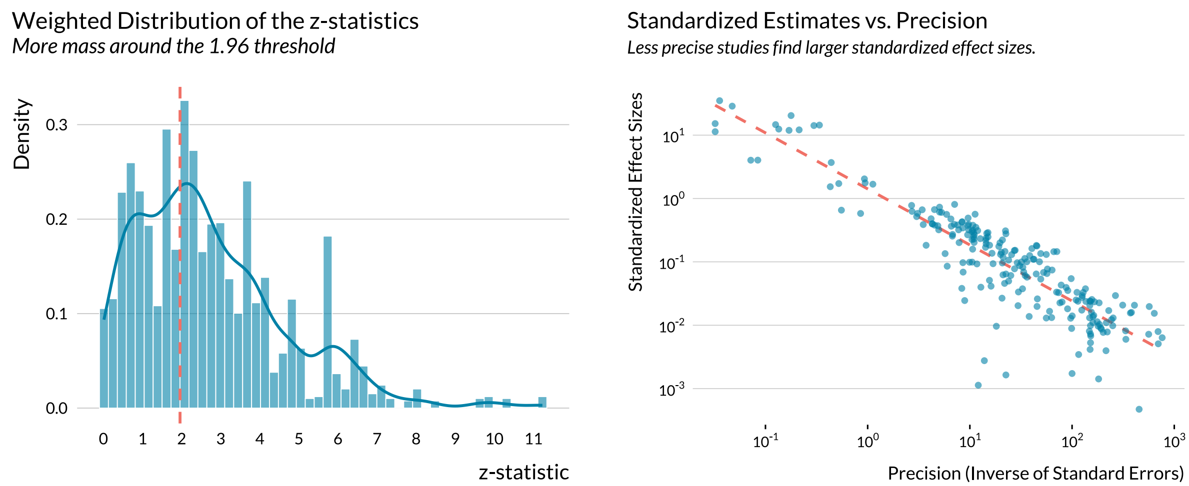
Figure 3: Distribution of Power and Exaggeration Ratio for Instrumental Variable Designs, Assuming that the Naive OLS Estimates Are the True Effect Sizes.



Notes: For 49 statistically significant 2SLS estimates, we define the true values of effect size as the corresponding OLS estimates. Each blue line represents either the estimated statistical power (%) or exaggeration factor of a study's result. Orange lines are the median of the two metrics. For visual clarity, we do not display three extreme exaggeration ratios.

Exaggeration of significant estimates is actually problematic when such results are favored for publication. Following Brodeur, Cook and Heyes (2020), we implement a caliper test. It hints to the presence of selection on significance in the causal literature. It formalizes the results displayed on the left panel of Figure 4: there is an excess mass in the weighted z-statistics distribution at the 5% statistical significance threshold. The right panel of Figure 4 produces further evidence of this favoring of significant estimates but also points to exaggeration. Imprecise designs yield larger standardized effect sizes. If published estimates captured true effects, their standardized effect size should be independent of the precision of the study. We are however not able to implement convincing heterogeneity analyses, considering the relatively small number of estimates considered. The method for identifying selective publication described in Andrews and Kasy (2019) also underlines the presence of a favoring of significant results and of exaggeration in the literature.

Figure 4: Suggestive Evidence of Publication Bias and Exaggeration in the Causal Inference Literature on Acute Health Effects of Air Pollution.



Notes: The sample in the left panel includes all 382 estimates reported in articles from the causal literature, excluding "naive" OLS estimates and placebo tests. Following Brodeur, Cook and Heyes (2020), the weights are equal to the inverse of the number of tests in the article. In the right panel we exclude the "naive" OLS estimates and placebo tests. Both axes are on a log10 scale. Limiting the sample to economics journals leaves the figures essentially unchanged (see supplemental material). Distinguishing between top 5 and other journals shows that even if their standardized effect sizes are typically smaller in top 5 journals, the same inverse relationship can be observed.

2.3 Exaggeration in the Standard Epidemiology Literature

Hundreds of papers have been published on the short-term health effects of air pollution in epidemiology, medicine and public health journals. A large fraction of these articles rely on Poisson generalized additive models, which allow for flexible adjustment for the temporal trend of health outcomes and for non-linear effects of weather parameters. This literature spans over 20 years and has replicated analyses in a large number of settings, providing crucial insights into the acute health effects of air pollution.

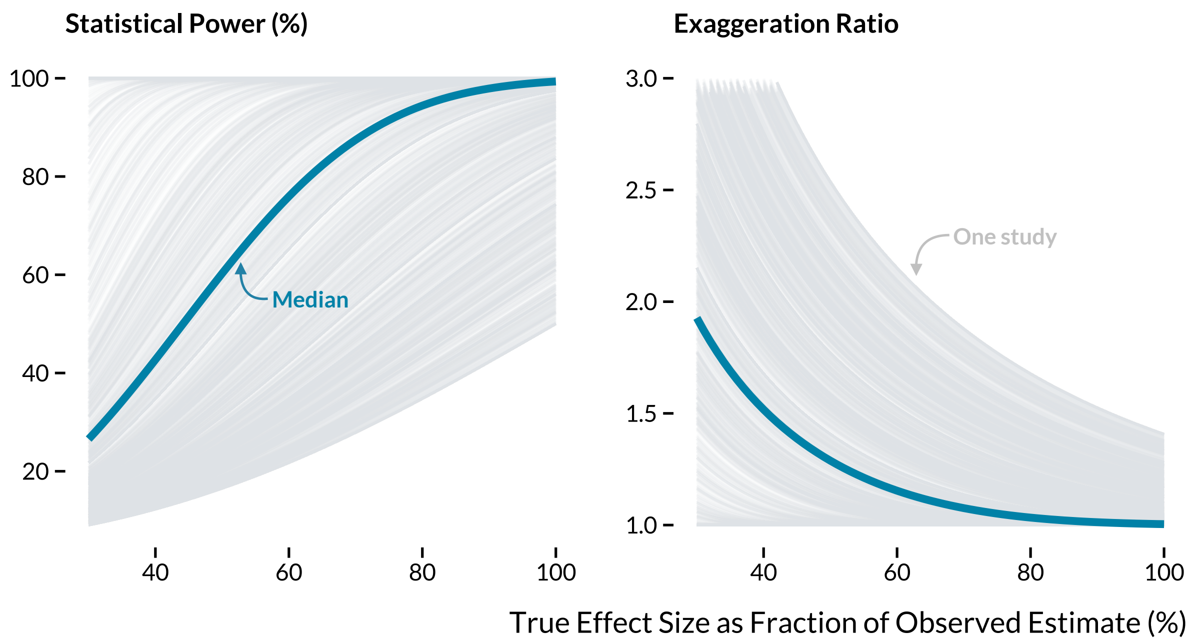
To gather a corpus of relevant articles, we run a search query on PubMed and Scopus combining air pollution terms with health outcomes, restricting to short-term effects. We retrieve the abstracts of 1834 articles. Conveniently, this literature routinely reports point estimates and confidence intervals in abstracts, allowing us to automatically extract results using regular expressions (RegEx). Our algorithm, available online, detects phrases such as "95% confidence interval (CI)" and retrieves the associated numbers. For instance, from the sentence: "`The excess risk for non-accidental mortality was` **`1.3% [95% CI, 0.8–1.7]`** `per 10` μ`g/m`^{3} `of PM10`" (Vichit-Vadakan, Vajanapoom and Ostro 2008), the algorithm recovers the estimate and its confidence interval (in blue). After manually reviewing abstracts and discarding off-topic articles, we retain 668 articles and 2,155 estimates.

Figure 5 displays the power and exaggeration curves for the 1,982 statistically significant estimates. If the true effect size was equal to half of the obtained estimate, 58% of studies would have a power below the conventional 80% threshold. The median exaggeration ratio would be 1.3, though this masks substantial heterogeneity: for one quarter of studies, the exaggeration ratio would exceed 1.9. This heterogeneity does not seem to be systematically correlated with observable factors, even though health science journals appear less prone to power issues than other journals and the few studies that consider air quality indexes as a pollutant seem to have higher power. Notably, the share of low-power studies has stagnated since the 2000s.

To complement this analysis, we use meta-analysis estimates from Shah et al. (2015) as more informed hypothetical effect sizes for a subset of 94 studies on the effects of several air pollutants on mortality and emergency admission for stroke. For each of these studies, we run retrospective power calculations to evaluate their ability to retrieve its corresponding meta-analysis estimate. We find that 63% have power below 80% and a median exaggeration ratio of 1.6. As shown in Figure 6, results vary considerably by pollutant, from a median exaggeration of 1 for $PM_{2.5}$ to 13.4 for O_3 (the median for O_3 is not displayed for visual clarity).

Unlike the causal inference corpus, we do not observe the full set of published estimates but only those reported in their abstracts. These are headline results that authors chose to emphasize and that are most likely to inform policymaking. As shown in Figure 7, there is striking bunching at the 1.96 threshold, indicating that statistical significance strongly influences which results are highlighted, constituting a specific form of selection on significance. Since identification approaches are relatively uniform across this literature, studies are less likely to be evaluated on methodological grounds, making the significance and magnitude of results all the more central to publication decisions.

Figure 5: Power and Exaggeration Curves for the Epidemiology Literature.



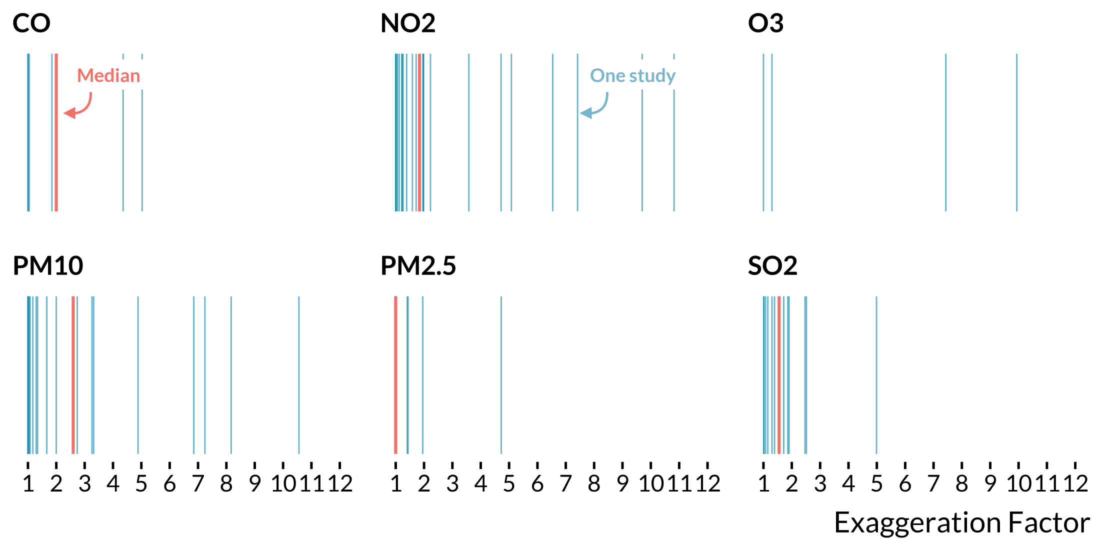
Notes: Each gray line is a power curve or an exaggeration curve of a statistically significant result published in the epidemiology literature. The blue lines are the median values. For visual clarity, we drop results for which exaggeration ratios were too large.

2.4 Impacts of Exaggeration on Public Policy

We now explore the policy implications of this exaggeration, starting with a description of the U.S. regulatory context. Under the Clean Air Act, systematic reviews of the scientific literature, synthesized in Integrated Science Assessments (ISAs), are used by the Environmental Protection Agency (EPA) to set National Ambient Air Quality Standards (NAAQS) (U.S. EPA 2015).[4] Beyond standard-setting, academic estimates enter the benefit cost analyses of major regulations through Regulatory Impact Analyses (RIAs), which compare the social costs and monetized benefits of alternative regulatory approaches (U.S. EPA 2024*a*). To quantify health benefits, the EPA relies on the Benefits Mapping and Analysis Program (BenMAP-CE), a modeling tool that combines air quality projections with concentration-response coefficients drawn from the epidemiology literature to estimate the economic value of changes in air pollution concentrations (Sacks et al. 2018). States are responsible for implementing these standards through State Implementation Plans (SIPs), and often draw on EPA-provided analytical tools such as BenMAP-CE, as well as guidance from RIAs and associated technical documents, to select compliance

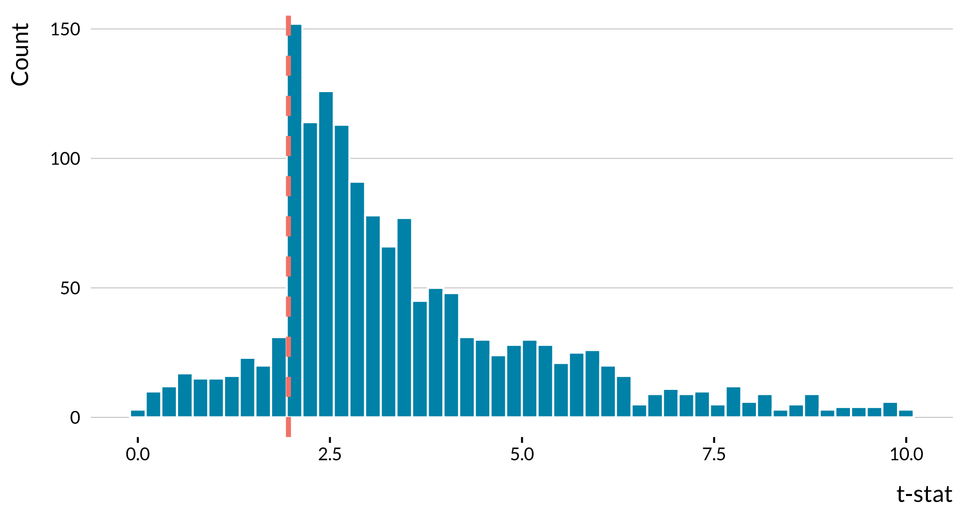
[4]This reflects the regulatory framework in place through the end of 2024.

Figure 6: Distribution of Exaggeration Ratios for Studies in Shah et al. (2015)'s Meta-Analysis.



Notes: Each blue line is the exaggeration ratio of a statistically significant estimate retrieved from Shah et al. (2015)'s meta-analysis. We use the meta-analysis estimates as true effect sizes in the retrospective power calculations. Orange lines are the medians. Extreme exaggeration ratios are removed for visual clarity. The median for O_3 is 13.4.

Figure 7: Distribution of t-statistics of Estimates in Abstracts from the Standard Epidemiology Literature



Notes: All estimates come from abstracts. Estimates with t-stat > 10 are not reported here for readability. The vertical dashed line represents the usual 1.96 threshold.

measures (U.S. EPA 2023).

In BenMAP-CE, health benefits from air quality improvements are computed in several steps (Sacks et al. 2018). After modeling changes in air pollution, the software relates them to changes in health outcomes using concentration-response functions derived from the epidemiological literature. These studies typically estimate log-linear concentration-response functions: $\Delta Y = \text{Pop} \times Y_0 \times (1 - e^{-\hat{\beta} \times \Delta AQ})$ where ΔY denotes the change in health outcomes attributed to air pollution, Pop the exposed population, Y_0 the baseline incidence, ΔAQ the specified change in air quality, and $\hat{\beta}$ the estimated health effect coefficient. For the small concentration changes typically considered in regulatory analysis, this simplifies to $\Delta Y \approx \text{Pop} \times Y_0 \times \hat{\beta} \times \Delta AQ$. The economic value of avoided health effects is then computed by multiplying ΔY by an estimate of the economic value of this health effect, *e.g.*, for mortality, the Value of a Statistical Life (VSL), whose central estimate recommended by the EPA is approximately \$10.7 million in 2022 dollars.[5] These monetized benefits can then be compared against abatement costs to compute the net benefits of alternative policies.

The monetized health benefits are proportional to the true health effects β. If the estimated health effects are exaggerated by a factor $\zeta = \frac{\hat{\beta}}{\beta}$, perceived benefits will be exaggerated by the same factor. To explore concrete implications on policy, consider the 2024 RIA for the revision of the $PM_{2.5}$ standards, which lowered the annual standard from 12 to 9 $\mu g/m^3$ (U.S. EPA 2024*a*). The RIA estimates annual monetized benefits in 2032 between \$22 billion and \$46 billion (2017\$) against annual costs of \$590 million. These health benefits are explicitly derived from a review of academic studies and calculated using BenMAP-CE. Given that the benefits of regulation outweigh costs by more than an order of magnitude, even substantial exaggeration leading to inflated perceived benefits would not eliminate positive net benefits. However, true net benefits would be lower than the regulator expects. Table 2 describes the impact of the exaggeration ratios estimated in Section 2 on the RIA benefit-costs analysis for the high expected benefit estimate of \$46 billion.

These discrepancies are economically large: annual misperceptions of net benefits range from \$10.5 to \$35.3 billion. Policy makers relying on exaggerated estimates would overstate the social surplus and the economic justification for stringent standards. When benefit-cost analyses are used to compare potential regulations, systematic exaggeration of health benefits could make air quality interventions appear more attractive than alternative health policies that may yield higher returns. The same logic applies when comparing different types of air pollution regulations or standard levels. If exaggeration varies across evaluations of these different regulations, due for instance to differences in

[5] EPA's central VSL estimate of \$4.8 million (1990\$) adjusts to approximately \$10.7 million with the Consumer Price Index (U.S. EPA 2024*b*).

Table 2: Policy Implications of Documented Exaggeration.

	Exaggeration ratio	Expected Net Benefits (B$/year)	Actual Net Benefits (B$/year)	Difference (B$/year)
Epidemiology median	1.3	45.4	34.9	10.5
Epidemiology 25th pctile	1.9	45.4	23.9	21.5
Causal inference median	1.7	45.4	26.7	18.7
Causal inference 25th pctile	2.0	45.4	22.7	22.7
IV vs. OLS median	4.5	45.4	10.1	35.3

Notes: Actual net benefits represent the net benefits under the hypothesis of exaggerated health effects: due to the linearity of the relationship between health effects and monetized benefits, they are simply computed as the expected net benefits of $45.4 billion (the high estimate for the 9 $\mu g/m^3$ standard) divided by the exaggeration ratio found in our analysis of the literature. The last column represents the difference between the expected and actual net benefits.

the populations affected, their ranking may change even when all options generate positive net benefits.

Air pollution regulations have benefit-cost ratios around 50, large enough that even substantial exaggeration leaves net benefits positive. In domains where estimated benefits and costs are closer in magnitude, exaggeration could have more dramatic implications: it could reverse the sign of a benefit-cost analysis and therefore determine whether a regulation is implemented.

3 Approach for the Simulation Analysis

While a literature review can provide evidence of exaggeration and lack of statistical power, it does not allow us to clearly identify the parameters driving it. We therefore implement a prospective analysis to overcome this limitation (Altoè et al. 2020, Black et al. 2022, for other examples of power simulations). We run Monte-Carlo simulations based on real data to emulate the main empirical strategies found in the literature. We use real data to avoid the difficult task of modeling the long-term and seasonal variations in health outcomes but also the specific effects of weather variables such as temperature. These simulations are grounded in the literature on the acute health effects of air pollution but nonetheless share many similarities with prevailing applied economics settings; panel data, large data sets, canonical identification strategies.

The present section describes how the simulations are implemented. Before that, we present the causal identification strategies used to measure the acute health effects of air pollution and then briefly describe the data used for the simulations.

3.1 Typical Research Designs to Measure the Short-Term Health Effects of Air Pollution

Several empirical strategies have been leveraged to estimate the short-term health effects of air pollution. We simulate the main ones existing in the literature. We consider the usual setting where data on air pollution, weather parameters, and health outcomes are at the daily-city level.

Standard regression approach. The standard strategy consists in directly estimating the dose-response between an air pollutant and a health outcome. In the epidemiology literature, researchers often rely on Poisson generalized additive models where they regress the daily count of a health outcome on an air pollutant concentration, while flexibly adjusting for weather parameters, seasonal and long-term variations. We approximate the workhorse model used by epidemiologists using linear models estimated via ordinary least squares:

$$\mathrm{Y}_{c,t} = \alpha + \beta \mathrm{P}_{c,t} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{1}$$

where c is the city index and t the daily time index. $\mathrm{Y}_{c,t}$ is the daily count of cases of a health outcome and $\mathrm{P}_{c,t}$ the average daily concentration of an air pollutant and $\epsilon_{c,t}$ an error term. The parameter β captures the short-term effect of an increase in the air pollutant concentration on the health outcome. To address confounding issues, the model adjusts for a set of weather covariates, $\mathbf{W}_{c,t}$, and calendar indicators $\mathbf{C}_t$.

Instrumental variable (IV) approach. The standard strategy could be prone to omitted variable bias and measurement error. A growing number of articles therefore exploits exogenous variations in air pollution. Most causal inference papers rely on IV designs where the concentration of an air pollutant is instrumented by thermal inversions (Arceo, Hanna and Oliva 2016), wind patterns (Schwartz, Fong and Zanobetti 2018, Deryugina et al. 2019), extreme natural events such as sandstorms or volcano eruptions (Ebenstein, Frank and Reingewertz 2015, Halliday, Lynham and de Paula 2019), or variations in transport traffic (Moretti and Neidell 2011, Knittel, Miller and Sanders 2016, Schlenker and Walker 2016). This approach can be summarized with a two-stage model where the first stage is:

$$\mathrm{P}_{c,t} = \delta + \theta \mathrm{Z}_{c,t} + \mathbf{W}_{c,t}\eta + \mathbf{C}_t\kappa + e_{c,t} \tag{2}$$

where $\mathrm{Z}_{c,t}$ is the instrumental variable. The second stage is then:

$$\mathrm{Y}_{c,t} = \alpha + \beta \widehat{\mathrm{P}}_{c,t} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{3}$$

where $\widehat{\mathrm{P}}_{c,t}$ is the exogenous variation in an air pollutant predicted by the instrument. The causal effect measured by this approach is a weighted average of per-unit causal responses to an increase in the concentration of an air pollutant (Angrist and Imbens 1995).

Reduced-form approach. A subset of articles directly estimates the relationship between the health outcome and exogenous shocks to air pollution. For instance, articles using this approach exploit public transport strikes or thermal inversion as exogenous shocks (Bauernschuster, Hener and Rainer 2017, Jans, Johansson and Nilsson 2018, Godzinski, Castillo et al. 2019, Giaccherini, Kopinska and Palma 2021). They estimate a model of the form:

$$\mathrm{Y}_{c,t} = \alpha + \beta \mathrm{D}_{c,t} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{4}$$

where $D_{c,t}$ is a dummy equal to 1 when city c is affected by a shock at time t and 0 otherwise. The parameter β captures an intention-to-treat effect.

Regression-discontinuity design (RDD) approach. Another empirical strategy found in the literature measures the effects of air quality alerts with a regression-discontinuity design (Chen et al. 2018, Anderson, Hyun and Lee 2022). In this approach, the following model is estimated for observations within an air pollution concentration bandwidth around the alert threshold:

$$\mathrm{Y}_{c,t} = \alpha + \beta \mathbf{1}\{P_{c,t} > P_c^{(a)}\} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{5}$$

where $P_c^{(a)}$ is the air pollution alert threshold for city c. We restrict our simulations to the case of sharp RDD. This model estimates the intention-to-treat effect of air quality alerts. It can both capture the effect of a subsequent decrease in air pollution caused by traffic restriction policies and inhabitants' avoidance behavior.

3.2 Data

The simulation exercises rely on a subset of the US National Morbidity, Mortality, and Air Pollution Study (NMMAPS). The dataset is publicly available and has been used in several major studies in the early 2000s to measure the short-term effects of ambient air pollutants on mortality outcomes (Peng and Dominici 2008). Specifically, we extract data at the city-day level for 68 cities over the 1987-1997 period. It corresponds to 4,018 daily observations per city, for a total sample size of 273,224 observations. We select observations on the average temperature (C°), the standardized concentration of carbon monoxide (CO), and mortality counts for several causes. We focus on CO as it is the air pollutant measured in most cities over the period and its concentration is strongly correlated to that of other pollutants such as particulate matter. Less than 5% of carbon monoxide concentrations and average temperature readings are missing in the initial data set. We impute them using the chained random forest algorithm implemented in the `missRanger` package (Mayer 2019).

3.3 Simulations Set-Up

General procedure. Our simulation procedure follows 7 main steps:

1. Randomly draw a study period and a sample of cities.
2. For instrumental variable, reduced-form and regression-discontinuity designs, randomly allocate days to exogenous shocks/air quality alerts.
3. Modify the health outcome, adding a treatment effect that we will try to recover.
4. Estimate the model.
5. Store the point estimate of interest and its standard error.
6. Repeat the procedure 1000 times.
7. Compute the proportion of statistically significant estimates at the 5% level (the power), the average of the absolute value of significant estimates over the true effect size (the exaggeration ratio), and the proportion of significant estimates of the opposite sign of the true effect (the probability to make a type S error).

Modeling assumptions. To only capture the specific issues arising due to low statistical power, we build our simulations such that (i) they meet all the required assumptions of empirical strategies and (ii) make it easier, compared to real settings, to recover the treatment effect. For all research designs, the treatment added to the data is not biased by unmeasured confounders nor measurement errors. For instrumental variable and reduced-form strategies, we only simulate binary and randomly allocated exogenous shocks (*e.g.* the occurrence of a thermal inversion). For the regression discontinuity approach, we only model sharp designs where an air quality alert is always activated above a randomly chosen threshold. The simulations always retrieve on average the true value of the treatment effect.

Two approaches for simulating research designs. For the reduced-form and regression discontinuity designs, we follow the Neyman-Rubin causal framework by simulating all potential outcomes (Rubin 1974). Consider that the health outcome value recorded in the NMMAPS dataset corresponds to the potential outcome $Y_{c,t}(0)$. To create the counterfactuals $Y_{c,t}(1)$, we add a treatment effect drawn from a Poisson distribution whose parameter corresponds to the magnitude of the treatment. We then randomly draw the treatment indicators $T_{t,c}$ for exogenous shocks or air quality alerts. For reduced-form strategies, the treatment status of each day is drawn from a Bernoulli distribution with parameter equal to the proportion of exogenous shocks desired. For air pollution alerts, we randomly draw a threshold from a uniform distribution and select a bandwidth such that it yields the desired proportion of treated observations. We finally express the observed values Y^{obs} of potential outcomes according to the treatment assignment: $Y^{obs}_{c,t} = (1\text{-}T_{c,t}) \times Y_{c,t}(0) + T_{c,t} \times Y_{c,t}(1)$.

To simulate standard regression and the instrumental variable strategies, we rely on a model-based approach. For the standard regression strategy, we first estimate the following statistical model on the data:

$$Y_{c,t} = \alpha + \beta Z_{c,t} + \mathbf{W}_{c,t}\lambda + \mathbf{C}_t\gamma + \epsilon_{c,t} \tag{6}$$

We then predict new observations of a $Y_{c,t}$ using the estimated coefficients of the model ($\hat{\beta}$, $\hat{\lambda}$, and $\hat{\gamma}$) and by adding noise drawn from a normal distribution with variance equal to that of the residuals $\widehat{\epsilon_{c,t}}$ (Peng, Dominici and Louis 2006). We modify the slope of the dose-response relationship by changing the value of the air pollution coefficient β. For the instrumental variable strategy, we use the same method as for the standard regression approach but first modify observed air pollutant concentrations $P_{c,t}$ according to the desired effect size θ of the randomly allocated instrument:

$$\widetilde{P}_{c,t} = P_{c,t} + \theta Z_{c,t} \tag{7}$$

We draw the allocation of each day to an exogenous shock from a Bernoulli distribution with parameter equal to the proportion of exogenous shocks. We then estimate a two-stage least squares model (2SLS) and modify the coefficient for the effect of the air pollutant on a health outcome. We finally generate the fake observations of the health outcome by combining the prediction from the modified 2SLS model and noise drawn from a normal distribution with variance equal to that of the residuals.

Varying parameters. To understand which parameters affect statistical power issues, we modify one aspect of the research design while keeping other parameters constant. We study the influence of four main parameters. First, we vary the sample size by drawing a different number of cities and changing the length of the study period. Second, we consider different effect sizes of air pollution or of an exogenous shock on the health outcome. Third, we allocate increasing proportions of exogenous shocks/air quality alerts. Fourth, we vary the number of cases in the outcome by considering different health outcomes.

Simulations of Case Studies. The simulations described above help explore the effect of each parameter on statistical power issues. Yet, the resulting set of parameters considered may not be perfectly representative of actual studies. To address this concern, we also calibrate simulation parameters to reproduce three papers published in the literature. We report these analyses in Appendix C.

4 Results of the Simulation Analysis

In this section, we describe how statistical power evolves with the treatment effect size, the number of observations, the proportion of exogenous shocks, the average count of the health outcome, and the strength of the instrument. In Appendix C, we show that statistical power issues can be substantial for actual parameter values found in the literature on acute health effects of air pollution.

4.1 Evolution of Power and Exaggeration with Study Parameters

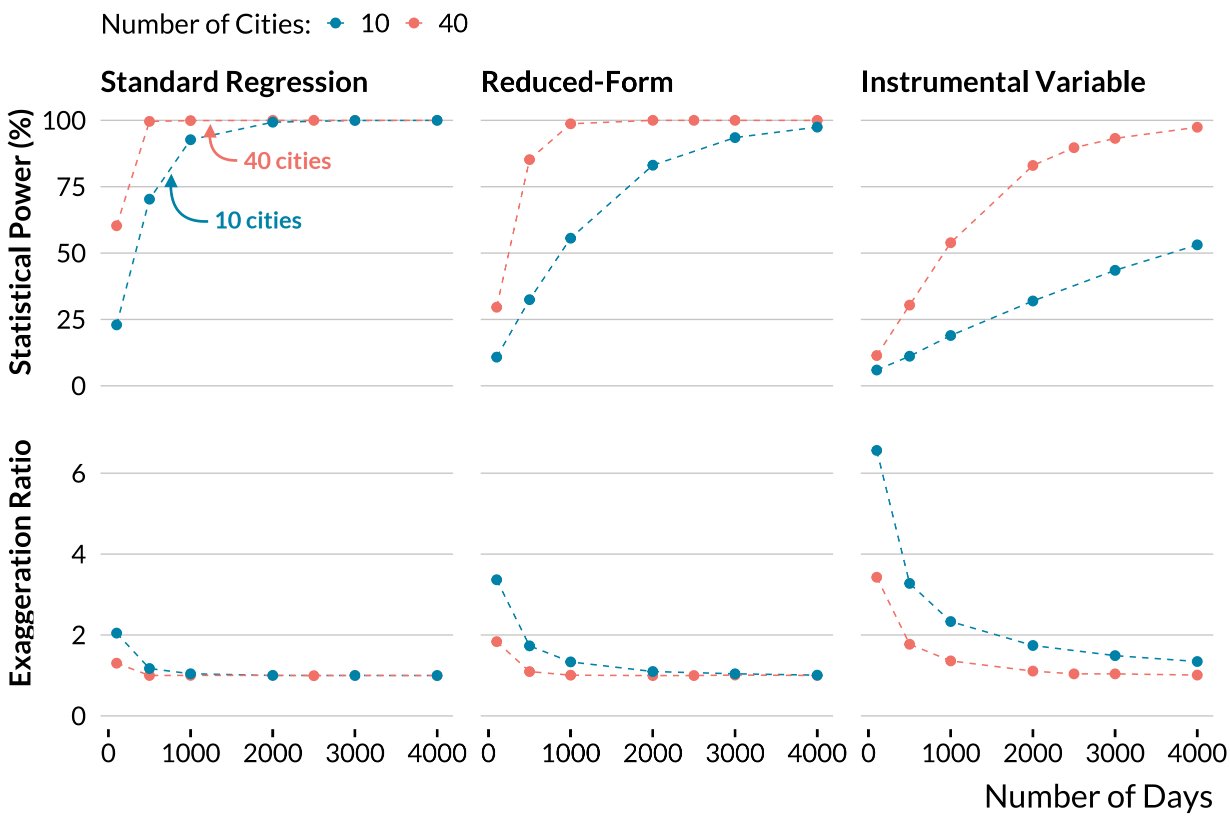
We aim to analyze how statistical power, exaggeration ratio and type S error are affected by the value of different study parameters. To do so, we set baseline values for these parameters and vary the value of each of them one by one. This enables us to get a sense of the impact of each parameter, other things held equal. We consider the following baseline parameters:

- A large sample size of 100,000 observations (2500 days $\times$ 40 cities),
- A 1% effect size, the order of magnitude found in the most precise studies of the literature. A one standard deviation increase in air pollution or a typical exogenous shock increases the health outcome by 1%,
- 50% of observations are subject to an exogenous shock. For air pollution alerts analyzed with regression discontinuity designs, we only consider observations close to the threshold, resulting in a smaller proportion of treated units: 10%,
- The health outcome is the total daily number of non-accidental deaths. It is the health outcome with the largest average number of counts (average daily mean of 23 cases).

For all statistical models, we adjust for temperature, temperature squared, city and calendar (weekday, month, year, month×year) fixed effects. We also repeat the simulations for a smaller sample size of 10,000 observations.

Sample Size

Figure 8: Evolution of Power and Exaggeration with Sample Size.

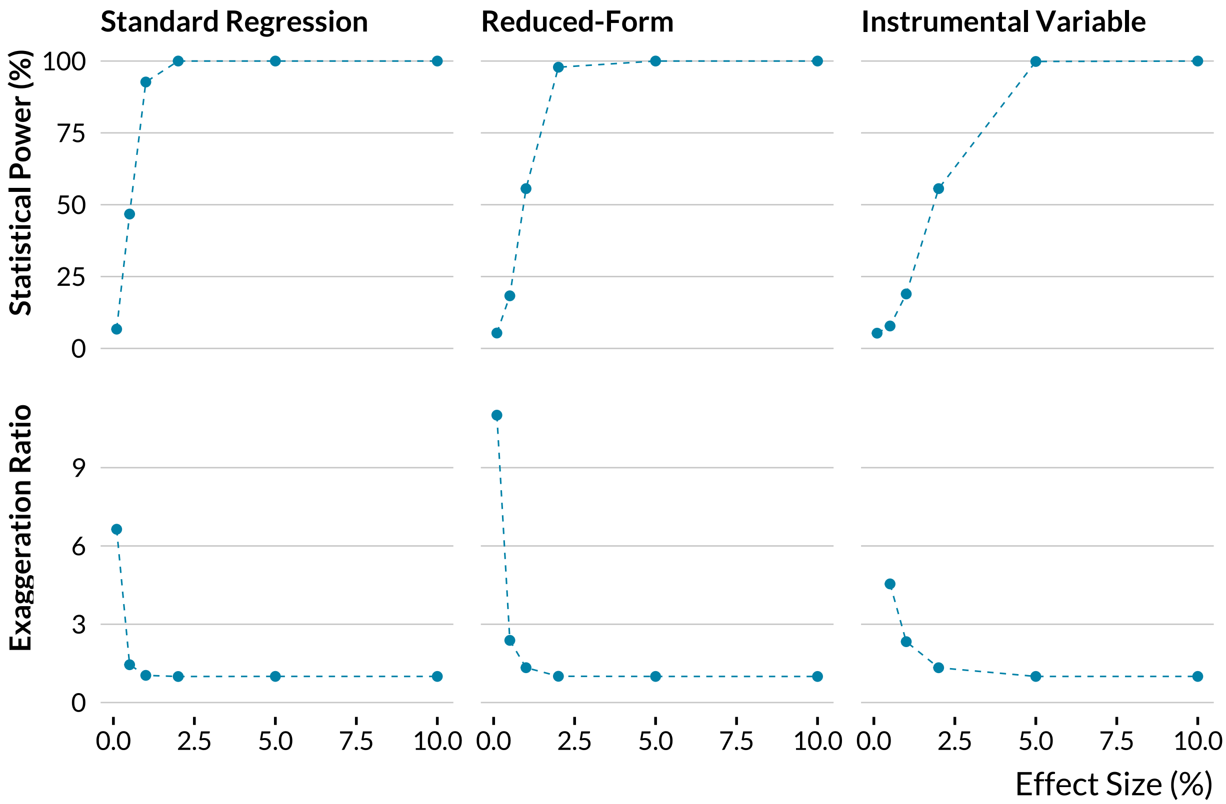


Notes: The other parameters are set to their baseline values: a true effect size of 1%, 50% of observations subject to an exogenous shock for instrumental variable and reduced-form designs, and the health outcome is the total number of non-accidental deaths.

We first find that statistical power and exaggeration issues can arise even for a large number of observations. For a sample size of 40,000 observations, the instrumental variable strategy only has a statistical power of 54% and exaggerates the true effect by a factor of 1.4, even with all other parameters set to conservative baseline values. As expected, Figure 8 also shows that the exaggeration ratio decreases and statistical power increases with the number of observations, as both quantities are driven by the variance of the estimator, which decreases with sample size (Zwet and Cator 2021, Lu, Qiu and Deng 2019). The standard regression strategy is much less prone to power issues than the instrumental variable strategy, as the variance of the two-stage least-squares estimator is larger than that of the ordinary least-squares estimator. In our simulations, the probability to make a Type S error is close to zero for all identification strategies and sample sizes.

Effect Size

Figure 9: Evolution of Power and Exaggeration with Effect Size.

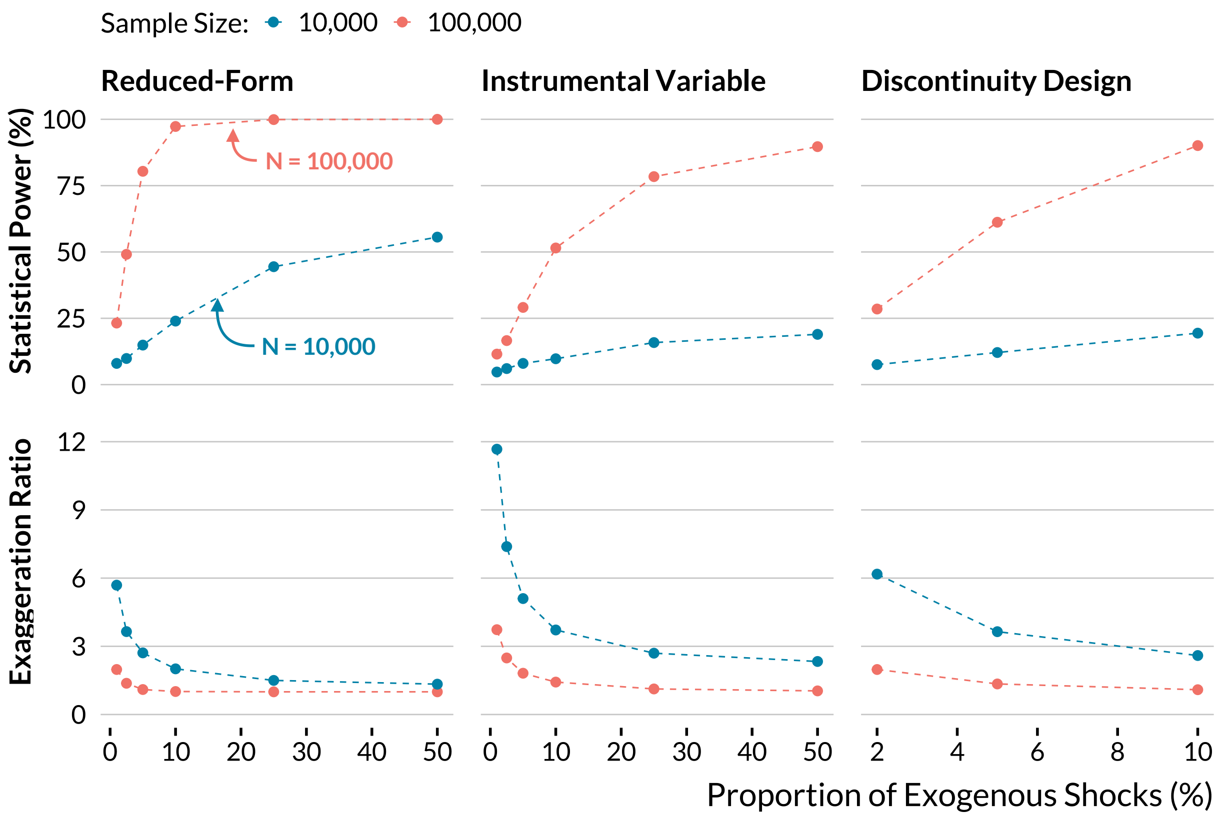


Notes: The sample size is 10,000. The other parameters are set to their baseline values: 50% of observations subject to an exogenous shock for instrumental variable and reduced-form designs, and the health outcome is the total number of non-accidental deaths. For an effect size of 1%, we do not display the exaggeration ratio of the instrumental variable design since it is above 20 and it would distort the graph.

Even with the large baseline sample size we consider, statistical power issues appear for effect sizes routinely found in the literature. For instance, for our instrumental variable strategy and an effect size of 0.5%, the average exaggeration ratio is about 1.7. As expected, Figure 9 also confirms that the exaggeration ratio decreases and statistical power increases with the true effect size (Zwet and Cator 2021, Lu, Qiu and Deng 2019). Standard regression and reduced-form strategies are less prone to power issues, even for small effect sizes.

Proportion of Exogenous Shocks

Figure 10: Evolution of Power and Exaggeration with the Proportion of Exogenous Shocks.



Notes: The other parameters are set to their baseline values: a true effect size of 1% and the health outcome is the total number of non-accidental deaths. The proportion of exogenous shocks corresponds to the fraction of days in the sample that are allocated to the treatment.

The link between the proportion of exogenous shocks and statistical power might be less widely known. In Figure 10, we show that statistical power increases with the proportion of treated units for instrumental variable, regression discontinuity and reduced-form designs. Conversely, the average exaggeration ratio increases as the proportion of exogenous shocks decreases.

This result can be explained by the fact that exaggeration increases and statistical power decreases with the variance of the estimator (Zwet and Cator 2021, Lu, Qiu and Deng 2019). Now, as routinely discussed for randomized controlled trials but seldom in the case of non-experimental studies, precision is maximized when half of the observations are exposed to the treatment of interest. The variance of the average treatment effect estimator (ATE) is $\sigma^2/[n \times p(1-p)]$ where σ is the standard deviation of the outcome in the treated and control groups and p the proportion of treated units. This quantity increases when p departs from 0.5. Thus, exaggeration increases when the proportion of exogenous shocks decreases, as long as it was initially smaller than 0.5.

Another way to interpret this result is to consider that a small number of exogenous shocks limits the variation that can be leveraged to identify the effect of interest. When the proportion of shocks decreases, the variance of the treatment variable decreases and therefore the variance of the estimator increases. A similar reasoning can be applied to IV strategies.

In practice, air pollution alerts, thermal inversion or transportation strikes are generally rare events. In some studies, they represent less than 5% of the observations. With a dataset of 10,000 observations, our simulations return an average exaggeration ratio of 2.7 for the reduced-form strategy. Despite large sample sizes, air pollution studies exploiting few exogenous shocks might be particularly prone to exaggeration issues.

Average Count of Cases of the Health Outcome

Table 3: Evolution of Power and Exaggeration with the Average Number of Daily Cases of Health Outcomes.

	Non-Accidental	Respiratory	COPD
Number of Cases	23	2	0.3
Statistical Power (%)	90	16	7.5
Exaggeration Ratio	1	2.4	5.9

Notes: This table displays the average number of cases, the power and the exaggeration ratio for three health outcomes: non-accidental deaths, respiratory deaths, and chronic pulmonary deaths for individuals aged between 65 and 75. These figures are obtained for the instrumental variable design with a sample size of 100,000 and 50% of observations subject to an exogenous shock. The instrument variable increases the air pollutant concentration by 0.5 standard deviation. A one standard deviation increase in the instrumented air pollutant leads to 1% relative increase in the health outcome considered.

Subgroup analyses are routinely carried out in the literature to evaluate the acute health effects of air pollution on children or the elderly. Yet, the average count of cases can also critically affect statistical power as shown in Table 3. For instance, in a setting with only a few deaths per day, a 1% increase in the number of deaths will rarely cause additional deaths. The effect will be more difficult to detect. To simulate situations with various numbers of cases, we consider three different outcome variables, with different counts of cases: the total number of non-accidental deaths (daily mean $\simeq 23$), the total number of respiratory deaths (daily mean $\simeq 2$) and the number of chronic obstructive pulmonary disease (COPD) cases for individuals aged between 65 and 75 (daily mean $\simeq 0.3$). Using baseline parameters and in the case of the large dataset, we find that statistical power is close to 100% for a 1% increase in the total number of non-accidental deaths. However, statistical power drops when the average count of cases decreases. For instance, the instrumental variable strategy has only 16% of statistical power to detect a 1% increase in respiratory deaths. The average exaggeration ratio is then equal to 2.4. For chronic obstructive pulmonary deaths—the health outcome with the lowest number of cases—the situation is even worse since the average exaggeration ratio reaches 5.9. When focusing on subgroups such as children or the elderly, one can expect to find larger effect sizes as those populations are more vulnerable to air pollution. While these larger effect sizes attenuate exaggeration concerns, the lower number of cases exacerbates them. It creates a trade-off for power issues.

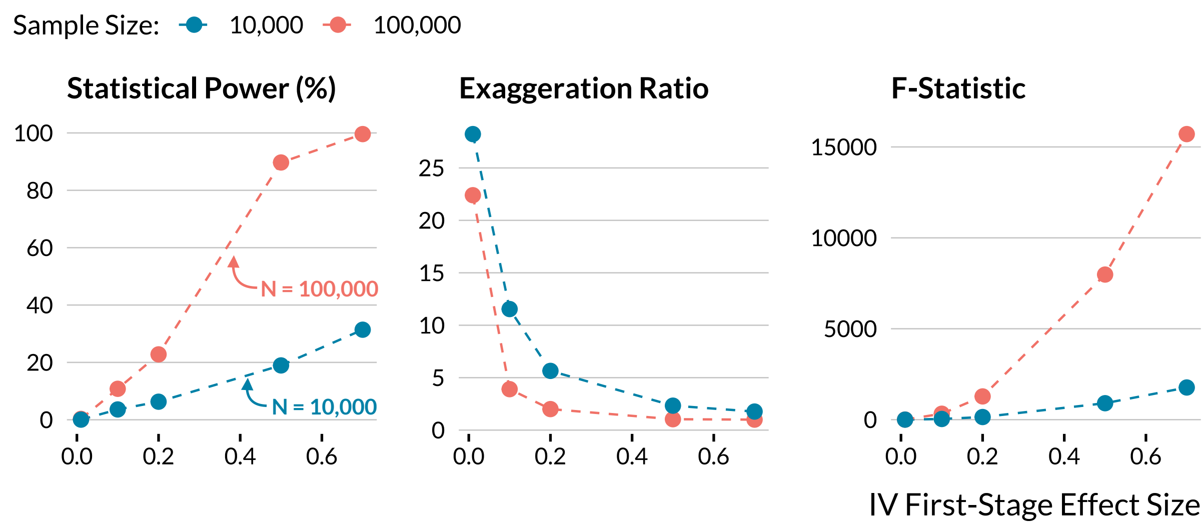
Issues Specific to the Instrumental Variable Design

In the case of instrumental variable strategies, statistical power is affected by the effect size in the first stage of the IV. In our simulations, we consider a binary instrument (e.g., the occurrence of a thermal inversion or a public transport strike). We define the effect size of the first stage of the IV as the standardized effect size of the instrument on the air pollutant concentration. An IV first stage effect size of 0.2 means that the instrument increases the concentration by 0.2 standard deviation.

As shown in Figure 11, we find that statistical power collapses and exaggeration soars when the IV first stage effect size decreases. Importantly, this issue even arises for large first-stage F-statistics. In our simulations based on the large data set with 100,000 observations, an instrumental variable's strength of 0.2, and an effect size of 1%, we find an average F-statistic of 1278. The statistical power is however only 23% and the average exaggeration ratio 2. A large F-statistic could therefore hide large exaggeration issues.

The relationship between IV strength and exaggeration comes from the fact that the variance of the 2SLS estimator decreases with the correlation between the instrument and the instrumented variable. In the homoskedastic case, the asymptotic variance of the 2SLS estimator is $\left(\mathbb{E}[XZ']\mathbb{E}[ZZ']^{-1}\mathbb{E}[ZX']\right)^{-1}\sigma^2$, where σ^2 is the variance of the error, X the endogenous variable and Z the instrument. When $\mathbb{E}[XZ']$ and $\mathbb{E}[ZX']$ decrease, the variance of the estimator increases. Again, since Zwet and Cator (2021) and Lu, Qiu and Deng (2019) show that as the variance of a normally distributed estimator increases, the statistical power decreases and exaggeration increases, we obtain the intuition for the simulation results.

Figure 11: Evolution of Power and Exaggeration with the Strength of the Instrumental Variable.



Notes: The true effect size is a 1% relative increase in the health outcome. The health outcome used in the simulations is the total number of non-accidental deaths. Half of the observations are exposed to exogenous shocks. The strength of the instrumental variable is defined as its effect in standard deviation on the air pollutant concentration.

5 Discussion

Growing evidence shows the existence of statistical power issues and publication bias towards statistical significance in economics, causing exaggeration (Brodeur et al. 2016, Ioannidis, Stanley and Doucouliagos 2017, Brodeur, Cook and Heyes 2020, Ferraro and Shukla 2020). Despite growing awareness of this problem, discussions about the drivers of exaggeration and therefore actionable guidance to tackle it in non-experimental economic research are still lacking (Altoè et al. 2020, Black et al. 2022). In this paper, we highlighted a list of concrete drivers of low power and exaggeration we should pay attention to when carrying out a non-experimental study.

While the simulations we ran were specific to studies on the acute health effect of air pollution, we argue that they can provide lessons for other types of non-experimental studies. First, a large literature investigates the short-term impacts of air pollution on different outcomes such as criminality, cognitive skills and productivity (Herrnstadt et al. 2021, Ebenstein, Lavy and Roth 2016, Adhvaryu, Kala and Nyshadham 2022, for instance). These studies use data with a very similar structure, only focusing on different outcomes and find effects of comparable magnitude or even smaller than those in our literature of interest. Our results should therefore be directly applicable to these literatures. More broadly, settings with typically low signal-to-noise ratios can by definition be subject to power and exaggeration issues. Since as described in Section 4, the impact of each driver we identified can be explained theoretically, we expect these drivers to affect power and exaggeration in other settings as well. A limited effect size, effective sample size, number of exogenous shocks, average count of the outcome or strength of the instrument can create exaggeration in many settings. Paying careful attention to these factors should help avoid exaggeration issues when running a non-experimental study.

In addition to these specific factors, when carrying out a study, we propose to systematically run retrospective calculations to gauge the risk of exaggeration. They are easy to implement and force us to discuss credible effect size. They allow evaluating if our research design enables us to confidently estimate a credible range of effect sizes. We implemented and discussed such calculations in our literature review and illustrate this approach in more detail in Appendix B by considering the example of Deryugina et al. (2019). As an even simpler first check, we also suggest to consider large confidence intervals verging 0 not only as a sign of uncertainty regarding the exact magnitude of the effect but also of limited power and potential exaggeration of the obtained point estimate.

Then, we advocate conducting prospective simulations before undertaking a non-experimental study. It allows verifying whether our design can detect effects of a credible magnitude in an almost-ideal setting. Unlike a retrospective analysis, it enables us to identify factors that drive exaggeration. Fake-data can be simulated from scratch or simulations can build on datasets used in other studies, as we did in the simulation section of this paper. To facilitate the adoption of this practice, we describe the template we use to run our simulations in the replication material. Black et al. (2022) also provide useful recommendations to implement power simulations.

More generally, we advocate paying attention to statistical power in non-experimental studies, even after a statistically significant estimate has been obtained, as insufficient power can lead to exaggeration and inaccurate published estimates. As such, we advocate reporting power calculations to demonstrate the robustness of the design and its ability to accurately capture smaller effect sizes.

On top of these specific comments, we should not forget that published estimates only suffer from exaggeration in the presence of publication bias. In its absence the resulting distribution would be centered around the true effect (Hernán 2022). Adopting a different view towards statistically insignificant results would in particular yield important benefits (Ziliak and McCloskey 2008, Wasserstein and Lazar 2016, McShane et al. 2019). To replace the null hypothesis testing framework, we advocate focusing on confidence intervals and to interpret the range of effect sizes supported by the data (Amrhein, Trafimow and Greenland 2019, Romer 2020). Qualifying estimates as "statistically significant" does not acknowledge the actual uncertainty that should be computed and embraced to better help policy-makers. Prospective and retrospective power analyses can help design better studies and improve the interpretation of their results.

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A List of Causal Inference Studies Considered

We display below studies included in the retrospective analysis of the causal inference literature. We group them by research designs:

Instrumental Variable Design: Moretti and Neidell (2011), Ebenstein, Frank and Reingewertz (2015), Schwartz et al. (2015), Arceo, Hanna and Oliva (2016), He, Fan and Zhou (2016), Knittel, Miller and Sanders (2016), Schlenker and Walker (2016), Sheldon and Sankaran (2017), Schwartz, Bind and Koutrakis (2017), Zhong, Cao and Wang (2017), Barwick et al. (2018), Hanlon (2018), Schwartz, Fong and Zanobetti (2018), Halliday, Lynham and de Paula (2019), Deryugina et al. (2019), Cheung, He and Pan (2020), Fan and Wang (2020), He, Liu and Zhou (2020), Giaccherini, Kopinska and Palma (2021), Godzinski and Suarez Castillo (2021), Guidetti, Pereda and Severnini (2021), Kim (2021), Liu and Ao (2021), Xia et al. (2022)

Reduced-Form Design: Bauernschuster, Hener and Rainer (2017), Jans, Johansson and Nilsson (2018), Jia and Ku (2019), Godzinski, Castillo et al. (2019)

Regression Discontinuity Design: Chen, Guo and Huang (2018), Fan, He and Zhou (2020), Anderson, Hyun and Lee (2022)

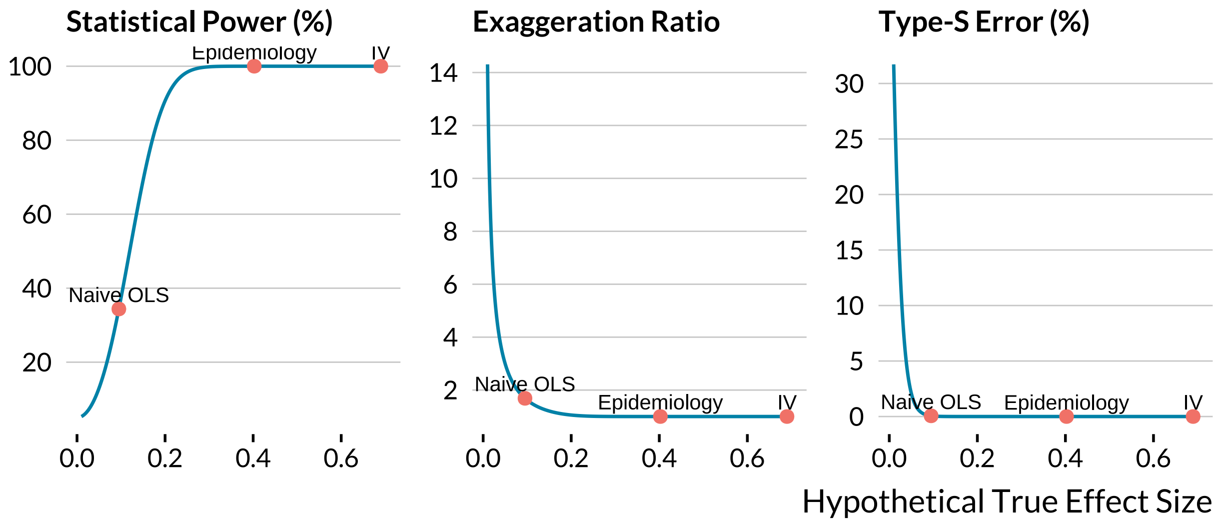
Event-Study Design: Mullins and Bharadwaj (2015), Simeonova et al. (2021)

Matching Design: Baccini et al. (2017), Forastiere, Carugno and Baccini (2020)

B Implementing a Retrospective Power Analysis

We explain here how we can easily implement a retrospective power analysis once a study is completed. In a flagship publication, Deryugina et al. (2019) instrument $PM_{2.5}$ concentrations with wind directions to estimate its effect on mortality, health care use, and medical costs among the US elderly. They gathered 1,980,549 daily observations at the county-level over the 1999–2013 period; it is one of the biggest sample sizes in the literature. When the authors instrument $PM_{2.5}$ with wind direction, they find that "a 1 μg/m^3 (about 10 percent of the mean) increase in $PM_{2.5}$ exposure for one day causes 0.69 additional deaths per million elderly individuals over the three-day window that spans the day of the increase and the following two days". The estimate's standard error is equal to 0.061. In Figure B.1, we plot the statistical power, the inflation factor of statistically significant estimates and the probability that they are of the wrong sign as a function of hypothetical true effect sizes.

Figure B.1: Power, Type M and S Errors Curves for Deryugina et al. (2019).



Notes: In each panel, a metric, such as the statistical power, the exaggeration ratio or the probability to make a type S error, is plotted against the range of hypothetical effect sizes. The "IV" label represents the value of the corresponding metric for an effect size equal to Deryugina et al. (2019)'s two-stage least square estimate. The "Epidemiology" label stands for the estimate found in Di et al. (2017), which is the epidemiology article most similar to Deryugina et al. (2019). The " Naive OLS" label corresponds to the estimate found by Deryugina et al. (2019) when the air pollutant is not instrumented.

The estimate found by Deryugina et al. (2019) represents a relative increase of 0.18% in mortality. We labeled it as "IV" in Figure B.1. Is this estimated effect size large compared to those reported in the standard epidemiology literature? We found a similar article to draw a comparison. Using a case-crossover design and conditional logistic regression, Di et al. (2017) find that a 1 $\mu g/m^3$ increase in $PM_{2.5}$ is associated with a 0.105% relative increase in all-cause mortality in the Medicare population from 2000 to 2012. The effect size found by Deryugina et al. (2019) is larger than this estimate labeled as "Epidemiology" in Figure B.1. If the estimate found by Di et al. (2017) was actually the true effect size of $PM_{2.5}$ on elderly mortality, the study of Deryugina et al. (2019) would have enough statistical power to perfectly avoid type M and S errors. Now, suppose that the true effect of the increase in $PM_{2.5}$ was 0.095 additional deaths per million elderly individuals—the estimate the authors found with a "naive" multivariate regression model. The statistical power would be 34%, the probability to make a type S error could be null but the exaggeration factor would be on average equal to 1.7. Even with a sample size of nearly 2 million observations, Deryugina et al. (2019) could make a non-negligible type M error if the true effect size was the naive ordinary least square estimate. Yet, the authors could argue that their instrumental variable strategy leads to a higher effect size as it overcomes unmeasured confounding bias and measurement error. Besides, for effect sizes down to 0.182 additional deaths per million elderly individuals (a 0.05% relative increase), their study has a very high statistical power and would not run into substantial type M error. A retrospective analysis is thus a very convenient way to think about the statistical power of a study to accurately detect alternative effect sizes.

C Case Studies

The main simulation results help understand how the various parameters influence the statistical power of studies. Yet, these parameters may not perfectly represent actual studies as we made several conservative assumptions: relatively large sample size, proportion of treated units, average outcome counts and instrumental variable strength. For each research design, we therefore consider a realistic set of parameters based on an example from the literature. We then vary the value of key parameters. As we are working with different data, we cannot exactly reproduce the level of precision found in the articles considered. Our goal is not to claim that the estimates produced by a particular article are inflated, but instead to understand how low power issues could arise for representative parameter values.

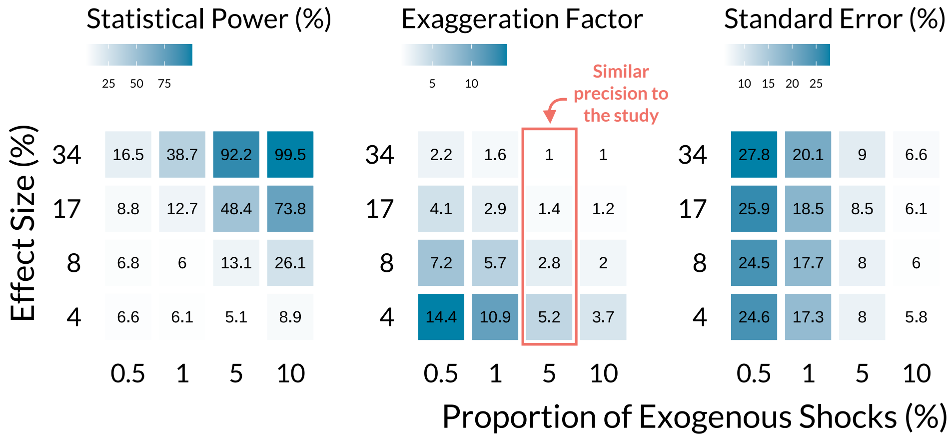
Public Transportation Strikes

Public transportation strikes are unique but rare positive shocks to air pollution as individuals use their cars to reach city centers. Even in a large data set, with several cities and a long study period, the proportion of affected days might be very small. For instance, Bauernschuster, Hener and Rainer (2017) investigate the effect of public transportation strikes on air pollution and emergency admission in the five biggest German cities over a period of 6 years. Despite a sample size of 11,000, there are only 57 1-day strikes during the study period (0.5% of days are actually treated). The authors find that children hospitalizations for breathing issues increase by 34% (SE=8%) on strike days. On average, 0.22 children per day go to the hospital for breathing issues.

We simulate a similar design with our own data. We first randomly sample 2200 observations for five cities and then vary (i) the proportion of exogenous shocks from 0.5% up to 10%, and (ii) the treatment effect size from a 4% increase up to a 34% increase. We focus on elderly mortality due to chronic obstructive pulmonary disease since it has an average daily count of 0.29 cases.

In Figure C.2, we display our simulation results. The first panel from the left shows that both large effect sizes and a large proportion of exogenous shocks are required to

Figure C.2: Evolution of Power and Exaggeration for Public Transportation Strikes Designs.



Notes: Each panel displays the average value of a metric (power, exaggeration, and standard error) for varying proportions of exogenous shocks and effect sizes. The average standard error of simulations is the raw standard error divided by the mean number of cases of the health outcome. For each combination of parameters, we ran 1000 simulations.

reach adequate power. In the middle panel, we show that a proportion of 0.5% of exogenous shocks is associated with very large exaggeration ratios, from 2.2 for a true effect size of 34% up to 14 for one of 4%. Power issues fade for a combination of a proportion of exogenous shocks above 5% and effect sizes above 17%. In the right panel, we plot the average standard error of the estimates, expressed as a fraction of the average of the health outcome. The standard error of Bauernschuster, Hener and Rainer (2017)'s is 8%. In our simulations, we recover that specific precision for a proportion of exogenous shocks of 5%. In that case, a true effect size of 34% would not yield inflated estimates. However, if effect sizes are actually smaller and more representative of those found in the literature, the exaggeration would be consequential.

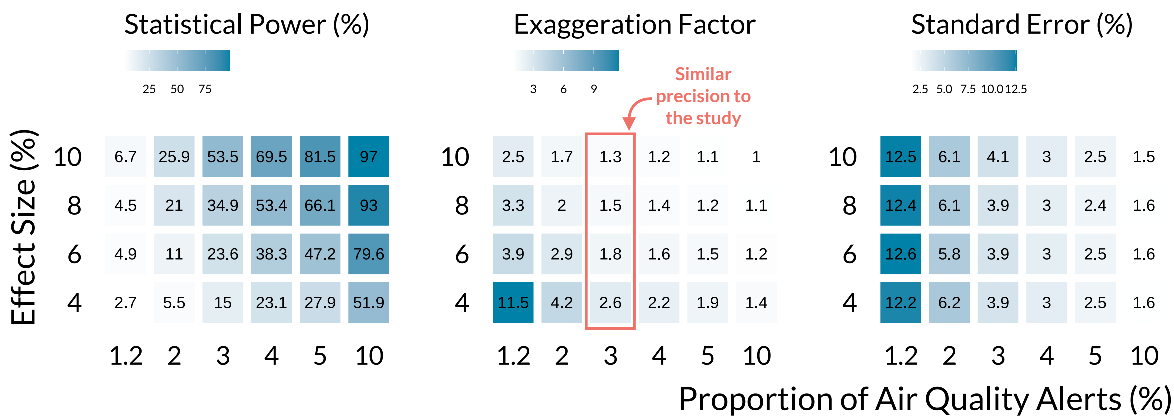
This simulation exercise shows that exaggeration is likely to arise in practice since the proportion of exogenous shocks is low. It occurs even when true effect sizes are relatively large.

Air Pollution Alerts

Air pollution alerts are also rare events. Their effects are estimated using regression discontinuity designs that restrict the analysis to observations close to the air quality threshold. As a consequence, the effective sample size may be particularly small. For instance, Chen et al. (2018) investigate the effects of air quality alerts on emergency department visits in Toronto, over the 2003-2012 period. While the nominal sample size is 3652, the effective one is only 143 (100 control days and 43 treated days). Only 1.2% of observations are treated. The authors find that eligibility to air quality (the intention-to-treat effect) approximately reduces emergency visits for asthma by 8% (SE=3.8%). The average daily count of cases of their health outcome is 26.

We approximate the setting of Chen et al. (2018) using our data. We first sample one city for a time period of 3652 days and randomly allocate the treatment. We then repeat the process varying the proportion of alerts and effect sizes. Our outcome variable is the total number of non-accidental deaths since it has a daily mean of 23.

Figure C.3: Evolution of Power and Exaggeration for Air Quality Alerts Designs.



Notes: Each panel displays the average value of a metric (power, exaggeration, and standard error) for varying proportions of exogenous shocks and effect sizes. The average standard error of simulations is the raw standard error divided by the mean number of cases of the health outcome. For each combination of parameters, we ran 1000 simulations.

Figure C.3 displays the simulations results. As in Figure C.2, a combination of large effect sizes and many air quality alerts is needed to avoid low power issues. We get a precision similar to Chen et al. (2018) for a proportion of air quality alerts of 3%. For an effect size of 4%, the average exaggeration ratio is equal to 2.6. In that case, the average of statistically significant estimates is 10%, which is similar to the effect size found by Chen, Guo and Huang (2018).

Unless true effect sizes are very large, air quality alert designs produce inflated estimates in realistic settings.

Instrumenting Air Pollution

Finally, we investigate the most commonly used strategy in the causal inference literature, the instrumental variable design. Several studies rely on very large datasets and exploit changes in weather patterns as sources of exogenous variations. For instance, Schwartz, Fong and Zanobetti (2018) instrument $PM_{2.5}$ concentration with planetary boundary layer, wind speed, and air pressure. Once the effects of seasonal and other weather parameters are accounted for, the combination of their instruments explains 18% of the variation in $PM_{2.5}$ concentration. They find that a 10 μg/m^3 increase in $PM_{2.5}$ leads to a 1.5% (SE=0.22%) increase in daily non-accidental mortality. There are on average 23 daily deaths in their dataset of 591,570 observations (135 cities with a length of study of approximately 4382 days).

In our simulations, we assess how the strength of the instrumental variable affects power issues for several health outcomes. We consider a binary instrumental variable and vary its effect on air pollution concentration from a 0.1 to a 0.5 standard deviation increase. The 18% correlation in Schwartz, Fong and Zanobetti (2018) corresponds to a 0.4 standard deviation increase in our case (Lipsey and Wilson 2001). We assume that half of the observations are exposed to exogenous shocks. We set an effect size corresponding to a 1.5% relative increase in three health outcomes with different average number of cases: non-accidental mortality (mean cases of 23), respiratory mortality (mean of 2), and chronic obstructive pulmonary mortality of elderly (mean of 0.3). Our data set being smaller than the one used in Schwartz, Fong and Zanobetti (2018), we only run simulations for a sample size of 100,000.

In Figure C.4, we see in the top-left panel that power reaches a satisfactory level for large instrumental variable strengths but only for non-accidental causes. For respiratory and elderly mortality, exaggeration can be substantial even for large IV strength. While our sample size is large, it is smaller than the one in Schwartz, Fong and Zanobetti (2018). As a consequence, our simulations only have a precision close to theirs for an instrumental variable strength of 0.5 and non-accidental mortality. Yet, our simulations highlight that important exaggeration issues can arise in realistic settings, even for large IV strength. The bottom-right panel of Figure C.4 confirms the result found in the simulations of the previous section: a large first stage F-statistic can be a poor indicator of statistical power issues. For instance, for non-accidental mortality and an IV strength of 0.1, the F-statistic is equal to 320 but the exaggeration factor is 2.6, with an associated estimated power of 16%. Importantly, as the F-statistic does not vary with the number of cases in the outcome it can all the more hide important power issues.

Figure C.4: Evolution of Power and Exaggeration for Instrumental Variable Designs.

Notes: Each panel displays the average value of a metric (power, exaggeration, standard error, and first-stage *F*-statistic.) for varying proportions of exogenous shocks and effect sizes. The average standard error of simulations is the raw standard error divided by the mean number of cases of the health outcome. For each combination of parameters, we ran 1000 simulations.